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## Assignment VI

### 1. Power of Quantum Register

At the time of writing, the whole world has less than  $10^{21}$  byte of memory. If we need two bytes to store one coefficient of a quantum state (assume one for the real part and one for the imaginary part), what is the maximum size of the quantum register state the world's memory can store?

### 2. Bell States

Prove all Bell states are entangled.

### 3. Profiting using Entanglement

There is a strange trillionaire living on Pluto. She invests based on the measurement of electron spin every month. If it is a spin up, she will invest in the gold on the Earth. If it is spin down, she will invest in silver on the Earth. An investor was able to entangle his electron with her electron to be measured next month secretly. Once the measurement is performed, the investor knows that his electron will collapse to the same spin states, and he will buy the corresponding commodity before anyone else who receives the result through a radio wave from Pluto. Is this really useful? In other words, is this a riskless investment for the investor on Earth?

### 4. Inner Product between Two Spaces

What is the inner product between the basis states of two electron spins? For example, what is the inner products of  $|0\rangle_1$  and  $|0\rangle_0$ ?

### 5. Entanglement Entropy of a Product State

Find the density matrix of  $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$ . Then find its entanglement entropy. (Hint: It should be zero because it is a product state.) Can you also factorize it into the product of two vectors in the  $\mathbb{C}^2$  space? Note that its density matrix is not diagonal. Try to diagonalize it first before taking the logarithmic values.

### 6. Partial Tracing

Find the entanglement entropy of  $|\phi^-\rangle$  but trace the  $B$  space. You should get the same answer. The equation now becomes

$$S_B = -T_{rA}(\rho_A \log_2 \rho_A) \text{ where } \rho_A = T_{rB}(\rho)$$

### 1. Power of Quantum Register

At the time of writing, the whole world has less than  $10^{21}$  byte of memory. If we need two bytes to store one coefficient of a quantum state (assume one for the real part and one for the imaginary part), what is the maximum size of the quantum register state the world's memory can store?

for  $n$ -qubit state  $2^n$  dimension space  
represent with complex number  $\rightarrow a + bi$   
2 parts (Real & Imag)  
each part for byte.  
 $\therefore$  need 2 byte.

$$\text{Memory needed} = 2^n \times 2 \text{ byte} \\ = 2^{n+1} \text{ byte}$$

$$\text{Max size of register} \quad 2^{n+1} \text{ byte} < 10^{21} \text{ byte}$$

$$2^n < 5 \times 10^{20}$$

$$n \ln(2) < \ln(5 \times 10^{20})$$

$$n < \log_2(5 \times 10^{20})$$

$$n < 68.76$$

$$\therefore n_{\max} = 68 \text{ qubits} \quad *$$

## 2. Bell States

Prove all Bell states are entangled.

Bell states:

$$|\phi^\pm\rangle = \frac{1}{\sqrt{2}} (|00\rangle \pm |11\rangle), \quad |\psi^\pm\rangle = \frac{1}{\sqrt{2}} (|01\rangle \pm |10\rangle)$$

If a state is not factorizable as  $|a\rangle \otimes |b\rangle$  then it is entangled.

$$\begin{aligned} \text{Suppose bell state equals } (\alpha|0\rangle + \beta|1\rangle) \otimes (\gamma|0\rangle + \delta|1\rangle) \\ = \alpha\gamma|00\rangle + \alpha\delta|01\rangle + \beta\gamma|10\rangle + \beta\delta|11\rangle \end{aligned}$$

For this to match  $|00\rangle + |11\rangle$  we need  $\alpha\delta = 0$ ,  $\beta\gamma = 0$   
but

$$\alpha\gamma \neq 0, \quad \beta\delta \neq 0$$

and the same way as  $|01\rangle + |10\rangle$

$\therefore$  Impossible  $\rightarrow$  Not separable  $\rightarrow$  Entangled. #Q.E.D

## 3. Profiting using Entanglement

There is a strange trillionaire living on Pluto. She invests based on the measurement of electron spin every month. If it is a spin up, she will invest in the gold on the Earth. If it is spin down, she will invest in silver on the Earth. An investor was able to entangle his electron with her electron to be measured next month secretly. Once the measurement is performed, the investor knows that his electron will collapse to the same spin states, and he will buy the corresponding commodity before anyone else who receives the result through a radio wave from Pluto. Is this really useful? In other words, is this a riskless investment for the investor on Earth?

Answer:

No, it's not riskless. Each local measurement is still random (50/50), so the investor can't know the outcome in advance. Entanglement only gives correlations, not faster-than-light information. To profit, they'd still need Pluto's classical signal, so there's no guaranteed advantage.

#### 4. Inner Product between Two Spaces

What is the inner product between the basis states of two electron spins? For example, what is the inner products of  $|0\rangle_1$  and  $|0\rangle_0$ ?

is the same qubit  $|0\rangle$  just different label

$$\therefore \langle 0|0 \rangle = 1 \quad \neq$$

#### 5. Entanglement Entropy of a Product State

Find the density matrix of  $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$ . Then find its entanglement entropy. (Hint: It should be zero because it is a product state.) Can you also factorize it into the product of two vectors in the  $\mathbb{C}^2$  space? Note that its density matrix is not diagonal. Try to diagonalize it first before taking the logarithmic values.

$$|\psi\rangle = \frac{1}{2} (|00\rangle + |01\rangle + |10\rangle + |11\rangle)$$

$$= \left( \frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) \otimes \left( \frac{|0\rangle + |1\rangle}{\sqrt{2}} \right)$$

$$= |+\rangle \otimes |+\rangle$$

in the basis  $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$

$$\rho = |\psi\rangle\langle\psi| = \frac{1}{4} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

$$\rho_A = \text{Tr}_B \rho = |+\rangle\langle+| = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

Diagonalize  $\rho_A$  in the  $\{|+\rangle, |-\rangle\}$  basis

$$\text{Eigenvalues: } \lambda_1 = 1, |+\rangle, \quad \lambda_2 = 0, |-\rangle$$

$$\text{Entropy: } S(\rho_A) = -\sum_i \lambda_i \log_2 \lambda_i = -(1 \cdot 0 + 0 \log_2 0) = 0 \quad \neq$$

$$|\psi\rangle = \left( \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right) \otimes \left( \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right) \text{ so it is a product state, consistent with our argument.}$$

## 6. Partial Tracing

Find the entanglement entropy of  $|\phi^-\rangle$  but trace the  $B$  space. You should get the same answer. The equation now becomes

$$S_B = -\text{Tr}_A(\rho_A \log_2 \rho_A) \text{ where } \rho_A = \text{Tr}_B(\rho)$$

$$|\phi^-\rangle = \frac{1}{\sqrt{2}} (|00\rangle - |11\rangle), \quad \rho = |\phi^-\rangle \langle \phi^-|$$

$$\text{partial trace over } B: \rho_A = \text{Tr}_B(\rho) = \frac{1}{2} (|0\rangle\langle 0| + |1\rangle\langle 1|) = \frac{I_2}{2}$$

$$\text{Eigenvalues of } \rho_A: \lambda_1 = \lambda_2 = \frac{1}{2}$$

$$S_B = -\text{Tr}_A(\rho_A \log_2 \rho_A) = -\sum_{i=1}^2 \frac{1}{2} \log_2 \frac{1}{2} = -2 \cdot \frac{1}{2} (-1) = 1 \quad \#$$

### **7. Unitarity of the NOT Gate**

Use hand calculation and co-lab to show that a 2-qubit NOT gate is unitary.

### **8. Unitarity of the CNOT Gate**

Use hand calculation and co-lab to show that the CNOT gate is unitary.

### **9. Implementation of a Quantum Circuit**

Use IBM-Q, by choosing “IBM Quantum Composer” to implement a 2-qubit circuit in which a 1-qubit NOT gate is applied to the MSB and then a CNOT gate is applied to the two qubits with the MSB as the control qubit. Refer to Assignment I, Problem 3 for how to construct a circuit. Firstly, based on what you have learned, what result do you expect? Add measurement to check if you get the desired result. Hint: since the input begins with  $|00\rangle$ , the output is 100%  $|11\rangle$ . And note that in IBM-Q, the bottom qubit is the MSB.

## 7. Unitarity of the NOT Gate

Use hand calculation and co-lab to show that a 2-qubit NOT gate is unitary.

Single qubit Not(x):  $X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ ,  $X^\dagger = X$ ,  $XX = I_2$

2-qubit Not(x):  $X \otimes X = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$

Unitarity:  $U^\dagger U = I$

$$\begin{aligned} & (X \otimes X)^\dagger (X \otimes X) \\ &= (X^\dagger \otimes X^\dagger) \otimes (X \otimes X) \\ &= (X^\dagger X) \otimes (X^\dagger X) \end{aligned}$$

$$= I_2 \otimes I_2$$

$$= I_4$$

∴  $X \otimes X$  is Unitary. ~~#~~

```
import numpy as np

X = np.array([[0, 1], [1, 0]])
U = np.kron(X, X)
print(np.allclose(U.conj().T @ U, np.eye(4)))
```

Problems Output Debug Console Terminal

```
• (quantum-computing) (base) pupipatsingkhorn
  /quantum-computing/assignment-06/q7.py
  True
○ (quantum-computing) (base) pupipatsingkhorn
```

## 8. Unitarity of the CNOT Gate

Use hand calculation and co-lab to show that the CNOT gate is unitary.

$$CNOT = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Column orthonormal  $\rightarrow$  Unitary

4 Cols are standard Basis vectors (a permutation of them), hence orthonormal

$$\rightarrow U^\dagger U = I$$

# Check

```
import numpy as np

CNOT = np.array([[1, 0, 0, 0], [0, 1, 0, 0], [0, 0, 0, 1], [0, 0, 1, 0]], dtype=complex)

# Check  $U^\dagger U = I$ 
print(np.allclose(CNOT.conj().T @ CNOT, np.eye(4)))
# -> True

# (Optional) verify action on basis
basis = np.eye(4)
print((CNOT @ basis).round(0))
```

```
• (quantum-computing) (base) pupipatsingkh
/quantum-computing/assignment-06/q8.py
True
[[1.+0.j 0.+0.j 0.+0.j 0.+0.j]
 [0.+0.j 1.+0.j 0.+0.j 0.+0.j]
 [0.+0.j 0.+0.j 0.+0.j 1.+0.j]
 [0.+0.j 0.+0.j 1.+0.j 0.+0.j]]
○ (quantum-computing) (base) pupipatsingkh
```



### 9. Implementation of a Quantum Circuit

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Since trial period of IBM Quantum Platform is expired, will use 'qiskit' instead.

Expect

$$|z_1 z_0\rangle = |00\rangle$$

$$\text{apply } X \text{ to MSB } (z_1) : |00\rangle \rightarrow |10\rangle$$

$$\text{apply CNOT (control } z_1, \text{ target } z_0) : \text{control}=1 \rightarrow \text{flip target } 0 \rightarrow 1 : |10\rangle \rightarrow |11\rangle$$

$$\text{Measure} \rightarrow 100\% |11\rangle$$

```
from qiskit import QuantumCircuit, transpile
from qiskit_aer import AerSimulator

qc = QuantumCircuit(2, 2)
qc.x(1) # NOT on MSB
qc.cx(1, 0) # CNOT
qc.measure([0, 1], [0, 1])

sim = AerSimulator()
compiled_circuit = transpile(qc, sim)
result = sim.run(compiled_circuit, shots=1000).result()
print(result.get_counts())
```

```
(datascience)
itories/quantu
{'11': 1000}
(datascience)
```